

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO2_AT3 — CASE STUDY

Course Code:	DSA06	Course Title:	Data Handling and Visualization
CO Assessed:	CO2 – Evaluation (BL5)	Assessment:	Assessment Tool 3 – Case Study
Weightage:	20% of CIA	Total Marks:	20 (2 Questions × 10 Marks)
CO2 Statement:	<i>Analyze real-world data scenarios and evaluate/justify the most appropriate chart types, coordinate systems, color scales, and dashboard designs to represent them.(BL5)</i>		
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Section / Batch:		Date:	

Code of Conduct

I Priyanka.G / 192524124 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Instructions

- This test contains 2 case-study questions, each carrying 10 marks. Total: 20 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 90 minutes.
- Each case study has four sub-parts (a–d); answer all parts for full marks.
- Support your answers with brief calculations or justifications where relevant.
- Diagrams may be hand-drawn or clearly described in words if drawing tools are unavailable.

Annexure A – CASE STUDY

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Case Study 1 Climate & Environmental Monitoring (10 Marks)

Scenario: An environmental research agency is preparing a report on regional climate and pollution trends. Their dataset includes: (i) a temperature anomaly reading for each monitored region, expressed as a deviation in °C from a historical baseline; (ii) a five-year Air Quality Index (AQI) trend, recorded monthly for a single city; and (iii) total annual carbon emissions broken down across five industrial sectors (Energy, Transport, Manufacturing, Agriculture, and Other).

Quick-Reference: Key Design Choices

a) Color Scale	Diverging (Red White Blue), centered at 0°C
b) AQI Trend Chart	Line chart with a 12-month rolling average overlay
c) Anomaly Map	Choropleth map / region×year grid heatmap + annotation
d) Emissions Chart	Tree map (preferred) vs. sector bar chart, both justified

a) What type of color scale is required for the temperature anomaly map, and why?

Answer:

A diverging color scale such as Red White Blue is required, not a sequential single-hue scale. Temperature anomaly is measured as a deviation from a historical baseline, so the data has a genuinely meaningful zero point (no change) at its center, and values can move in two qualitatively different directions from that center: positive anomalies (regions that have warmed) and negative anomalies (regions that have cooled). A diverging scale places a neutral color (white) at zero and lets two different hues grow in saturation as values move away from zero in either direction red for warming, blue for cooling so the sign of the change is encoded by hue and the magnitude of the change is encoded by saturation. A sequential scale cannot do this: it only has one hue running from light to dark, so it cannot distinguish a region that cooled by 3°C from one that warmed by 3°C, which would misrepresent the very relationship (warming vs. cooling) the map is meant to communicate.

192524124_ Global Temperature Anomaly (Filled Map)

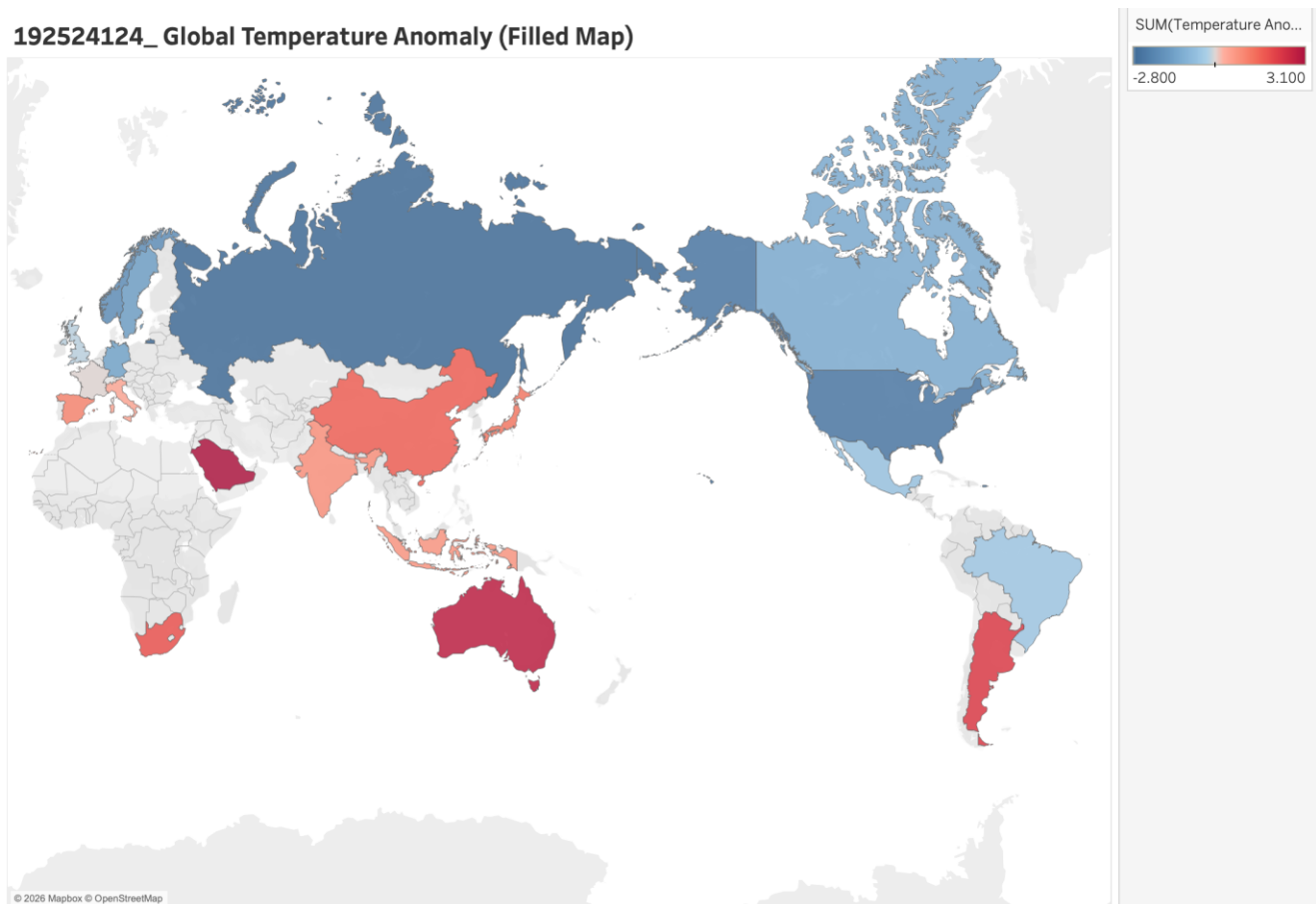


Figure 1a: Global Temperature Anomaly Diverging Blue White–Red Color Scale (Range: -2.8 to $+3.1$)

b) Propose a chart for the 5-year AQI trend and explain how you handle large seasonal swings in the data.

Answer:

A line chart is the appropriate choice, with time (month, spanning the full five years) on the X-axis and AQI value on the Y-axis, since AQI is a continuous numeric measurement recorded at regular intervals and a line chart is the standard way to reveal a trend's shape and direction over a continuous time axis.

Because air quality typically swings sharply with the seasons (e.g., worse in winter due to heating and inversions, better in monsoon/rainy periods), plotting five years of raw monthly values alone would produce a highly jagged, saw-tooth line in which the underlying multi-year trend is hidden behind repeating seasonal noise. To handle this, the raw monthly line should be layered with a second, smoothed series with a 12-month rolling (moving) average plotted as a bolder overlay line on top of the lighter raw data. The rolling average cancels the within-year seasonal cycle by construction (each point already averages a full year), so its shape reveals whether AQI is genuinely improving or worsening across the five years, independent of the time of year.

Illustrative calculation: suppose the first six months of raw monthly AQI readings for the city are as follows (values are for demonstration only, not the case's actual data):

Month	Jan	Feb	Mar	Apr	May	Jun
AQI (raw)	165	172	140	130	110	95

Illustrative sample AQI readings used to demonstrate the rolling-average technique

A 3-month rolling average in March is calculated as $(\text{Jan} + \text{Feb} + \text{Mar}) \div 3 = (165 + 172 + 140) \div 3 = 477 \div 3 = 159$. The window then slides forward by one month: the value in April becomes $(\text{Feb} + \text{Mar} + \text{Apr}) \div 3 = (172 + 140 + 130) \div 3 = 442 \div 3 \approx 147.3$, and at May it becomes $(\text{Mar} + \text{Apr} + \text{May}) \div 3 = (140 + 130 + 110) \div 3 = 380 \div 3 \approx 126.7$. Even though the raw values jump around considerably (e.g., a 32-point drop from Feb to Mar), the rolling-average series descends smoothly ($159 \rightarrow 147.3 \rightarrow 126.7$), which is exactly the kind of clear, de-noised downward trend a five-year, 12-month rolling average would reveal at a larger scale letting the report state with confidence whether air quality is genuinely improving year over year, rather than the reader having to mentally filter out seasonal noise from a raw monthly line themselves.

As a secondary technique, a shaded band (min–max range per calendar month across all five years) could be added behind the lines to show the seasonal envelope explicitly, letting a viewer see both the long-term trend (via the rolling average) and how much of the variation is 'normal' seasonal swing versus a genuine anomaly.

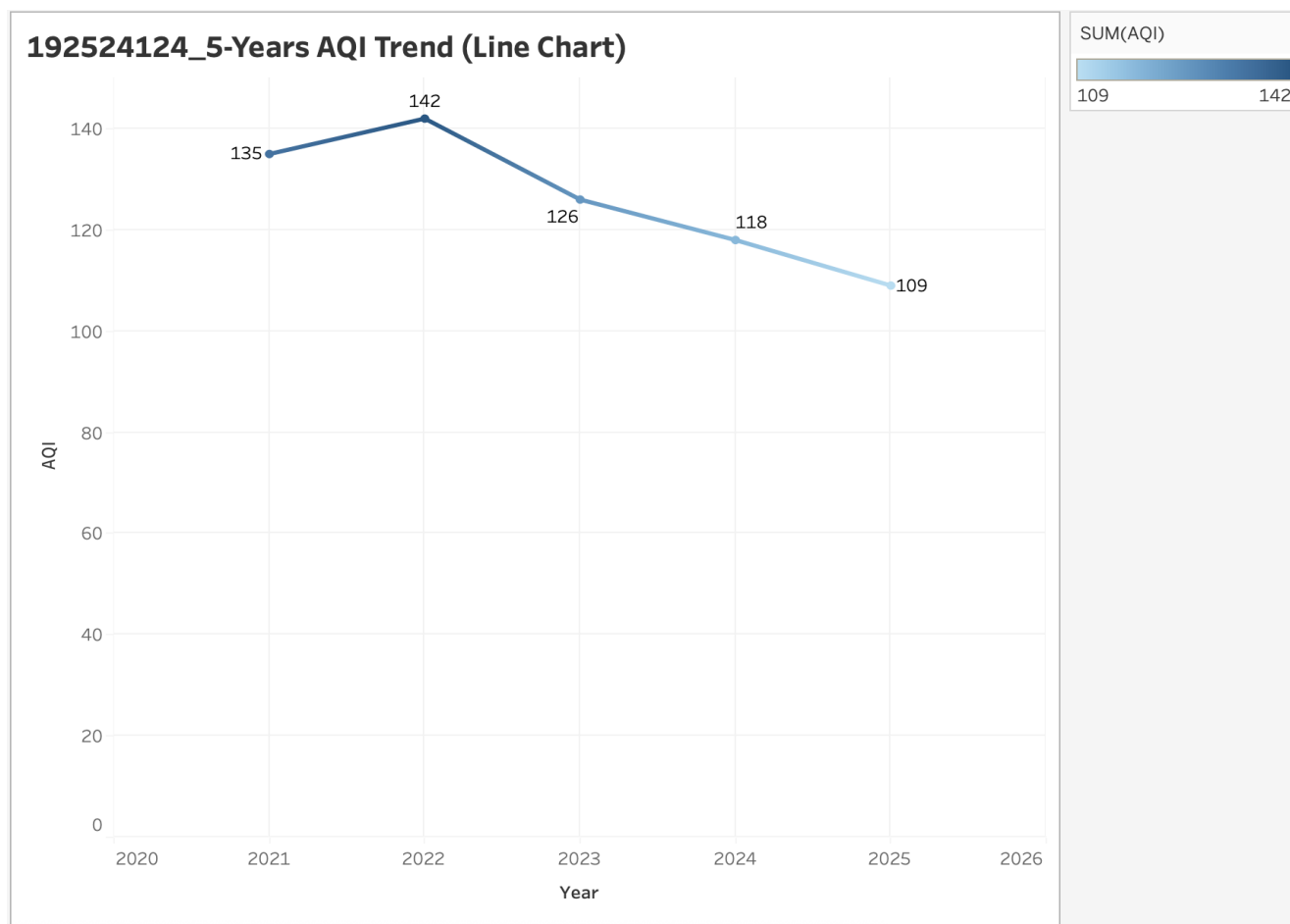


Figure 1b: 5-Year AQI Trend Annual Values (135 → 142 → 126 → 118 → 109)

c) Design a choropleth-style map or grid for the temperature anomaly data, combined with an annotation on the most extreme region.

Answer:

The primary design is a choropleth map: each monitored region is drawn as a filled geographic shape, and its fill color is set using the diverging Red White Blue scale from part (a), so the map communicates, at a glance, which regions have warmed (red), stayed near baseline (white), or cooled (blue), and by how much (saturation). If precise geographic boundary data is not available, an equivalent region-by-year grid (a matrix heatmap with regions as rows and years as columns, each cell colored on the same diverging scale) achieves the same encoding without requiring map geometry and has the added benefit of showing each region's anomaly trend across time rather than a single snapshot.

An annotation a callout box with a connecting line or arrow is placed directly on the region (or grid cell) with the largest positive anomaly, labelling it explicitly, e.g., "Most Extreme Region: [Region Name] +4.2°C above baseline, the highest anomaly recorded," so that the single most decision-relevant fact on the map does not have to be inferred by comparing color saturation across every region by eye.

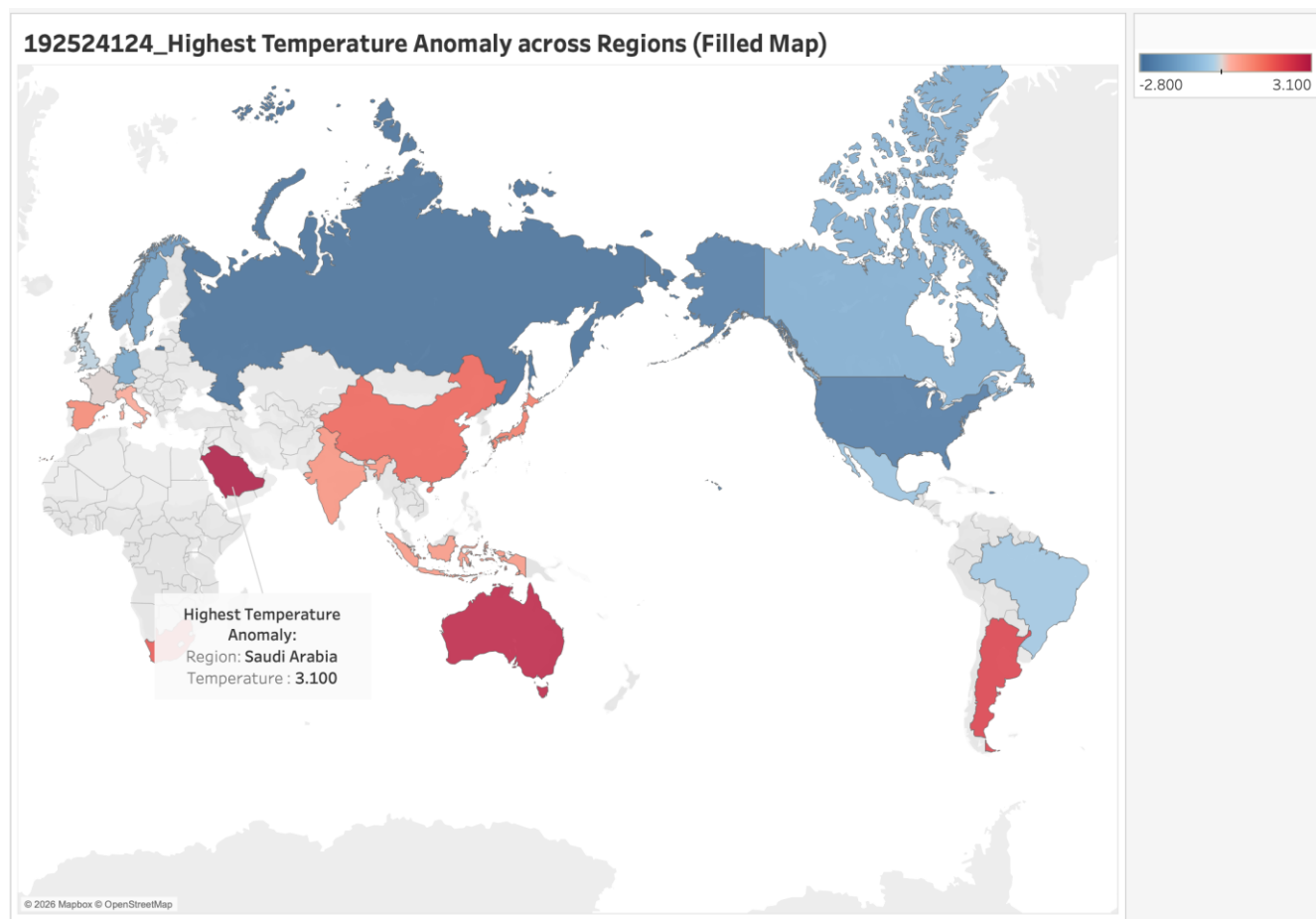


Figure 1c: Highest Temperature Anomaly Across Regions Annotated Choropleth (Saudi Arabia, +3.1)

d) Justify your choice of chart for carbon emissions by sector and explain why a tree map might be preferred over a pie chart for this particular breakdown.

Answer:

For a precise, side-by-side comparison of exactly five sectors, a sorted horizontal bar chart is generally the most accurate choice, because comparing bar lengths against a common baseline is a task human perception handles far more accurately than comparing angles or areas the classic finding behind Cleveland and McGill's hierarchy of elementary perceptual tasks.

Using the values shown in the constructed tree map (Figure 1d):

Sector	Energy	Transport	Industry	Agriculture	Residential
Emissions (kt)	420	250	180	90	60

Sector emissions as plotted in the constructed tree map (Figure 1d)

The total across all sectors is $420 + 250 + 180 + 90 + 60 = 1,000$ kt, so each sector's share of the whole is: Energy = 42%, Transport = 25%, Industry = 18%, Agriculture = 9%, and Residential = 6%. On a bar chart, these five percentages would appear as five bars of clearly different, easily ranked lengths. On a pie chart, however, the two smallest slices (Agriculture at 9% and Residential at 6%) would occupy only about 32° and 22° of the circle respectively thin enough that their labels would likely need to be pulled outside the pie with leader lines, and distinguishing a 9% slice from, say, an 11% slice by eye alone is difficult. The tree main Figure 1d instead renders Energy as the largest rectangle (proportional to 42% of the total area), down to Residential as the smallest, and if any sector is later subdivided (e.g., Energy into Coal/Gas/Renewables), each of those sub-values simply becomes a nested rectangle within the Energy block a hierarchical breakdown a pie chart has no mechanism to show at all.

However, if the report later needs to break sectors down further into sub-categories (e.g., Energy split into Coal, Gas, Renewables; Transport split into Road, Air, Rail) or needs to emphasize each sector's share of the total whole rather than just its raw magnitude, a tree map becomes the stronger choice over a pie chart. A tree map encodes each category's value as the area of a rectangle, tiled to fill the available space with no wasted whitespace, and critically can nest rectangles hierarchically, so a sector's rectangle can itself be subdivided into its constituent sub-sectors, something a flat pie chart cannot represent at all. A pie chart also struggles as category count grows past five or six slices (exactly the sector count here, on the edge of readability), because thin slices become hard to compare by angle and their labels have to be pulled outside the chart; a tree map's rectangles, by contrast, stay legible even with more categories, since text can be placed directly inside each rectangle as long as it is reasonably sized. For this specific five-sector, potentially hierarchical emissions breakdown, a tree map is therefore preferred over a pie chart, while a simple bar chart remains the better tool if the sole goal is the most precise flat comparison of five totals.

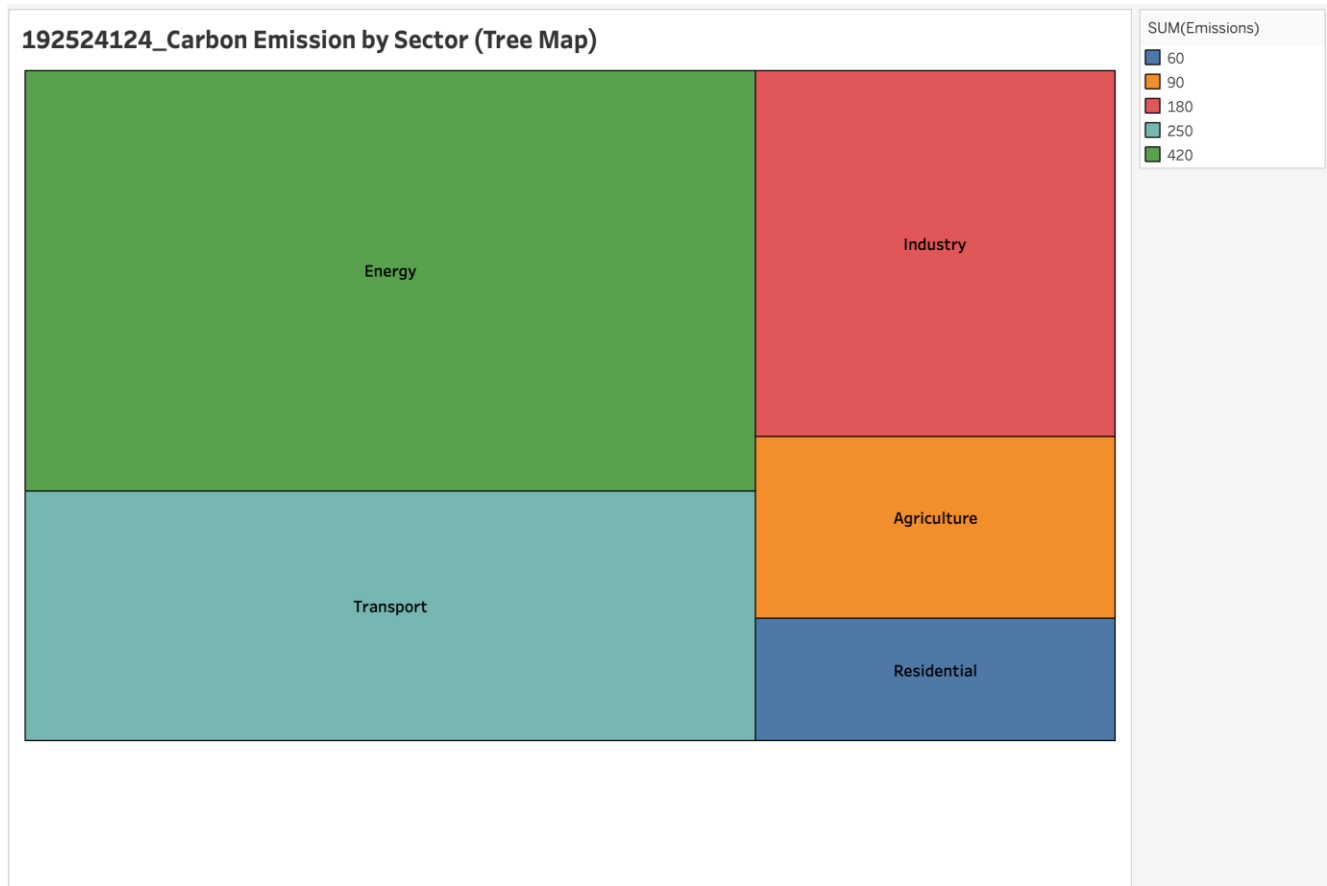


Figure 1d: Carbon Emissions by Sector Tree map (Energy 420 → Transport 250 → Industry 180 → Agriculture 90 → Residential 60)

Case Study 2 Municipal Budget & Service Satisfaction (10 Marks)

Scenario: A city's finance department is preparing a report for the city council covering: (i) total annual spending for each of the city's districts; (ii) residentsatisfaction with municipal services, collected as an ordinal rating per district (e.g., Very Dissatisfied, Dissatisfied, Neutral, Satisfied, Very Satisfied); and (iii) each district's year-over-year (YoY) percentage change in spending compared to the previous year, which may be positive (increase) or negative (decrease).

Quick-Reference: Key Design Choices

a) Axis Scale	Linear numeric scale, starting at zero (spending is a ratio quantity)
b) Satisfaction Chart	Ordered (ordinal) bar chart with a fixed rank-order axis and sequential/diverging color
c) YoY Change Chart	Diverging bar chart, Blue White–Red diverging color scale
d) Combined Dashboard	District spending-change + satisfaction, shared District axis, annotated + narrative

a) What axis scale is appropriate for the district spending data, and why?

Answer:

A linear numeric scale starting at zero is appropriate for the district's spending values. Spending is a ratio-type quantity – a district that spent twice as much as another should be represented by a bar twice as tall – and a bar chart can only communicate that ratio truthfully if its axis starts at zero; if the axis were truncated to start above zero, the visual ratio of the bar heights would no longer match the real ratio of the underlying values, which is especially important in a public accountability context such as a city council report, where readers will reasonably expect that a bar twice as tall represents genuinely double the spending.

Using the values shown in the constructed chart (Figure 2a):

District	South	Central	East	North	West
Spending	180	160	140	120	95

District spending as plotted in the constructed bar chart (Figure 2a)

The true ratio between the highest-spending district (South, 180) and the lowest (West, 95) is $180 \div 95 \approx 1.89$, meaning South genuinely spent roughly 1.9 times as much as West. On the correctly zero-based chart in Figure 2a, South's bar is drawn about 1.9 times taller than West's, matching this reality. If the axis were instead truncated to start at, say, 80, the two bars would only span 100 and 15 units respectively above that new baseline – a ratio of roughly 6.7:1 that would visually exaggerate the true ~1.9:1 difference more than three-fold. This is precisely why the zero-based axis used in Figure 2a is the correct choice for this kind of ratio comparison.

If district spending values happen to be extremely skewed (a few very large districts, many very small ones), a square-root or log scale may be considered as a secondary view but only with very explicit axis labelling, since a non-linear scale on its own can otherwise mislead a non-technical council audience; the default, safest choice remains a zero-based linear scale.

192524124_District Spending (Bar Chart)

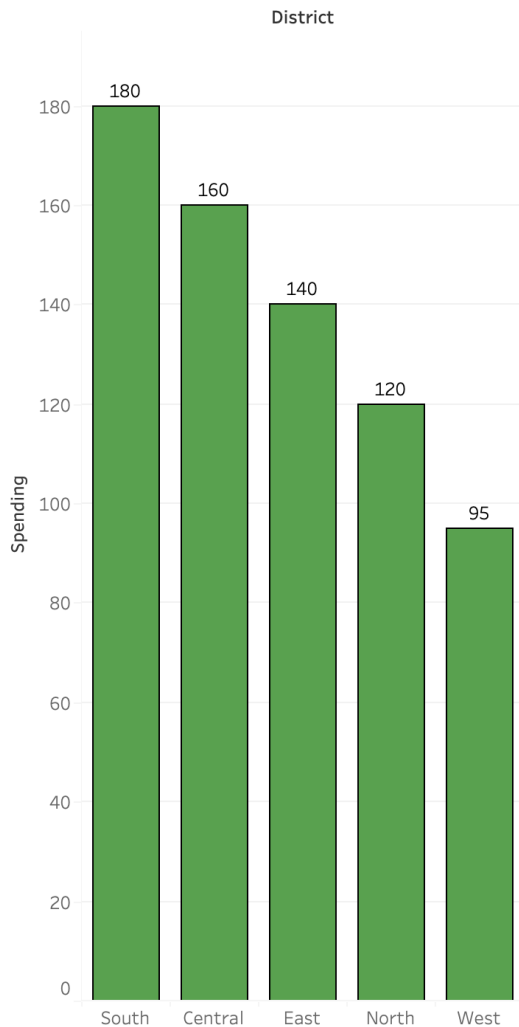


Figure 2a: District Spending Zero-Based Linear Bar Chart

b) Which chart correctly represents the ordinal service-satisfaction categories while preserving their order?

Answer:

An ordered bar chart (or an ordered horizontal bar/Likert-style chart) correctly represents this data, provided two conditions are met: first, the five satisfaction categories (Very Dissatisfied → Dissatisfied → Neutral → Satisfied → Very Satisfied) must be placed along the axis in their natural rank order, not resorted alphabetically or by frequency, since ordinal data's defining property is that its categories have a meaningful sequence that must be visually preserved; second, if color is used to distinguish the categories, it should be a sequential or diverging color ramp (e.g., a single hue growing darker from 'Very Dissatisfied' to 'Very Satisfied', or a diverging red-to-green scale centered on 'Neutral') rather than an arbitrary set of unrelated qualitative colors, because an unordered color palette would visually imply the categories are unrelated nominal groups rather than points along a single satisfaction scale. For comparing satisfaction across multiple districts at once, a diverging stacked bar chart (each district's bar split into segments for each rating, with negative-

leaning ratings extending left of a center line and positive-leaning ratings extending right) is a strong extension of this same ordinal-preserving principle.

The constructed chart (Figure 2b) follows exactly this design: the five categories stay in their correct rank order along the X-axis, and the fill color moves from red (Very Unsatisfied) through orange and yellow to teal and green (Very Satisfied), so the color itself reinforces the ordinal sequence rather than contradicting it. Reading the bar heights (Very Unsatisfied = 35, Unsatisfied = 58, Neutral = 96, Satisfied = 148, Very Satisfied = 120) shows that a clear majority of the 457 total responses (268, or roughly 59%) fall in the 'Satisfied' or 'Very Satisfied' categories, though the 96 'Neutral' and 58 'Unsatisfied' responses together represent a meaningful minority worth monitoring.

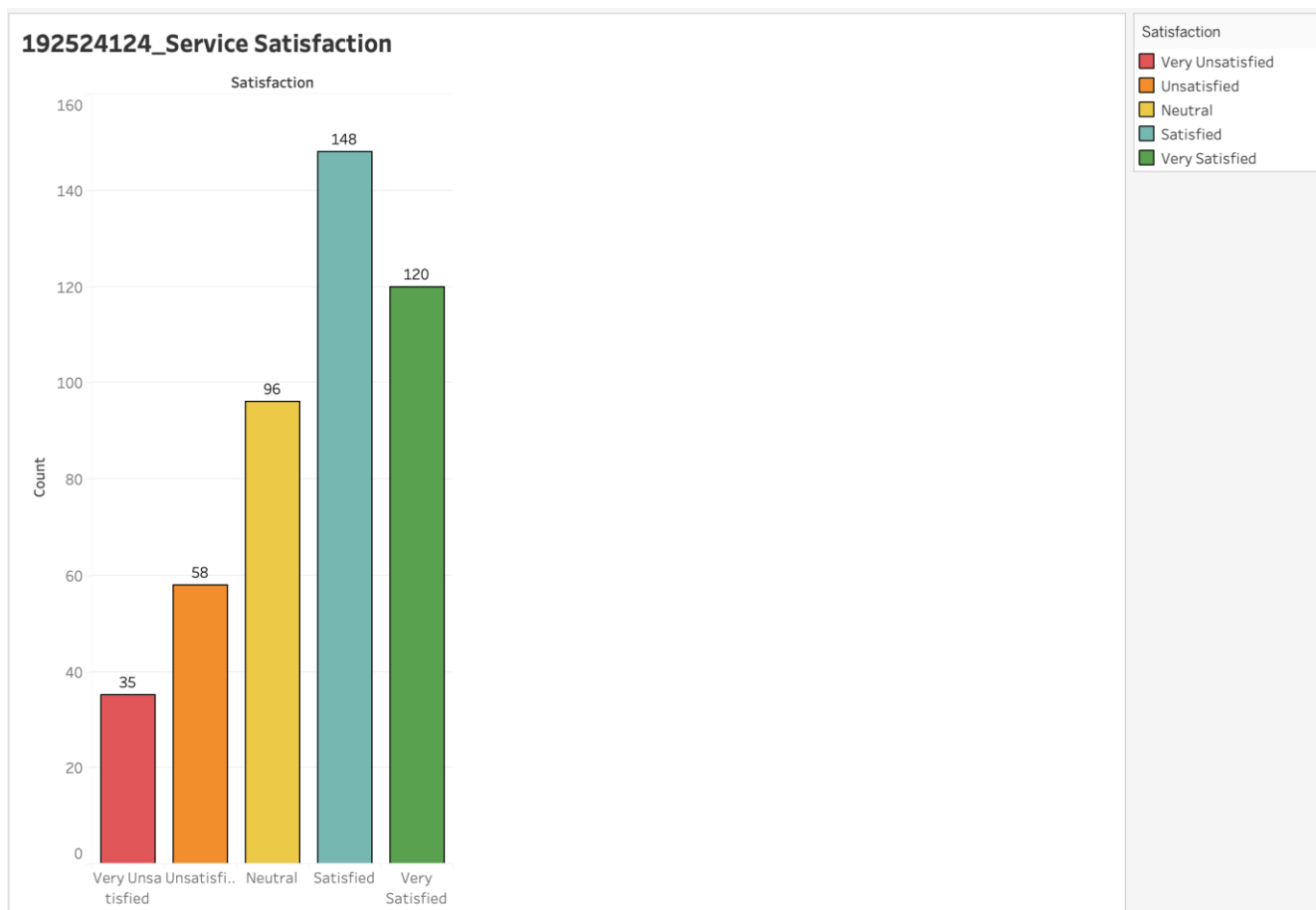


Figure 2b: Service Satisfaction Ordered Bar Chart Preserving Rank Order

c) Propose a chart and color scale for the year-over-year spending change data (increases vs. decreases).

Answer:

A diverging bar chart is proposed: one bar per period, extending above a central zero line for periods where spending increased, and below that zero line for periods where spending decreased, with bar length proportional to the magnitude of the percentage change. This encoding is used because YoY spending change has a meaningful, non-arbitrary zero point (no change from the previous year) and moves in two genuinely different directions from it

mirroring the same reasoning used for the temperature anomaly map in Case Study 1: only a diverging encoding can correctly show both the direction (increase or decrease) and the magnitude (how much) of the change in a single chart.

Using the values shown in the constructed chart (Figure 2c), which plots this change year by year for the period 2021–2025:

Year	2021	2022	2023	2024	2025
Spending Change (%)	+15	−8	+22	−5	+18

Year-over-year spending change as plotted in the constructed diverging bar chart (Figure 2c)

2021 (+15%), 2023 (+22%), and 2025 (+18%) each draw a bar extending above the zero line, with 2023's bar the tallest of the three, since a 22% increase is the single largest change in either direction across the five years shown. 2022 (−8%) and 2024 (−5%) each draw a bar extending below the line; both comparatively modest cuts next to the size of the increases.

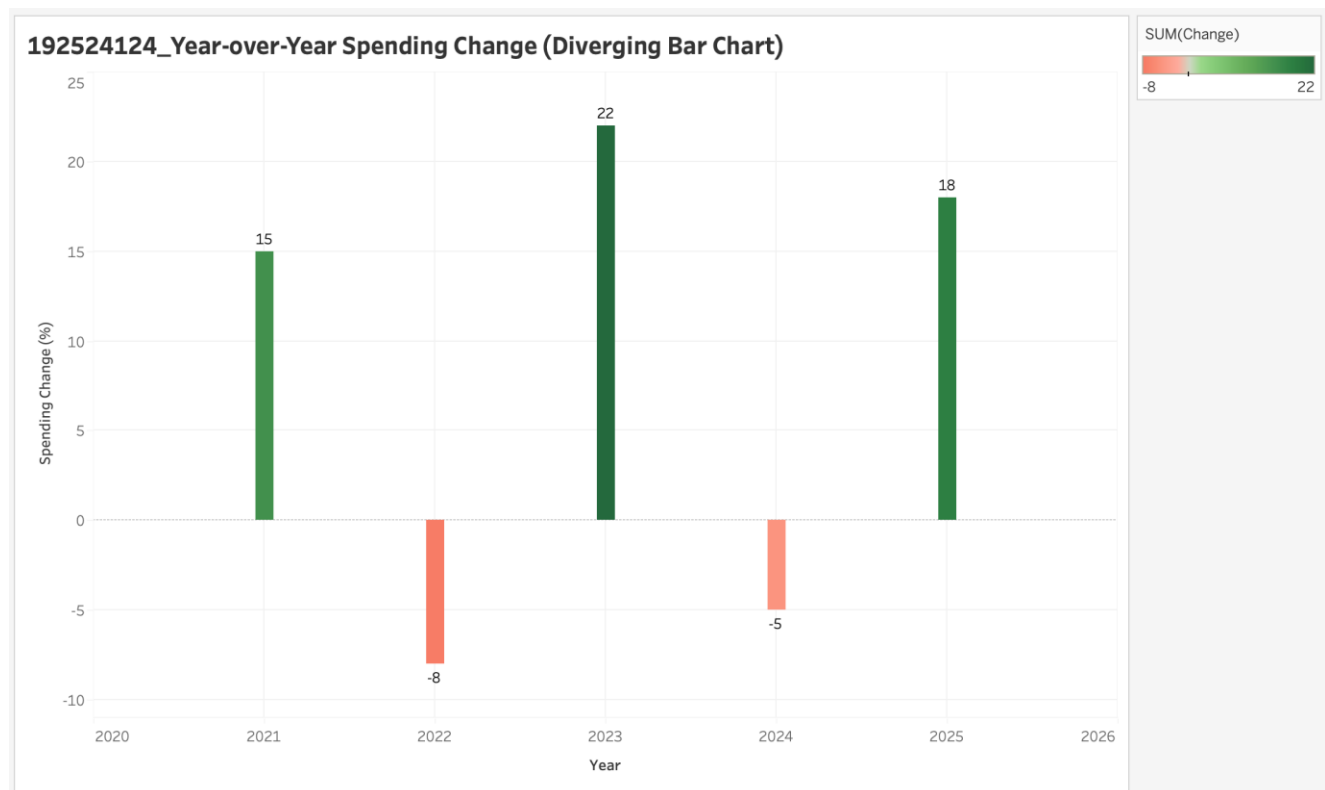


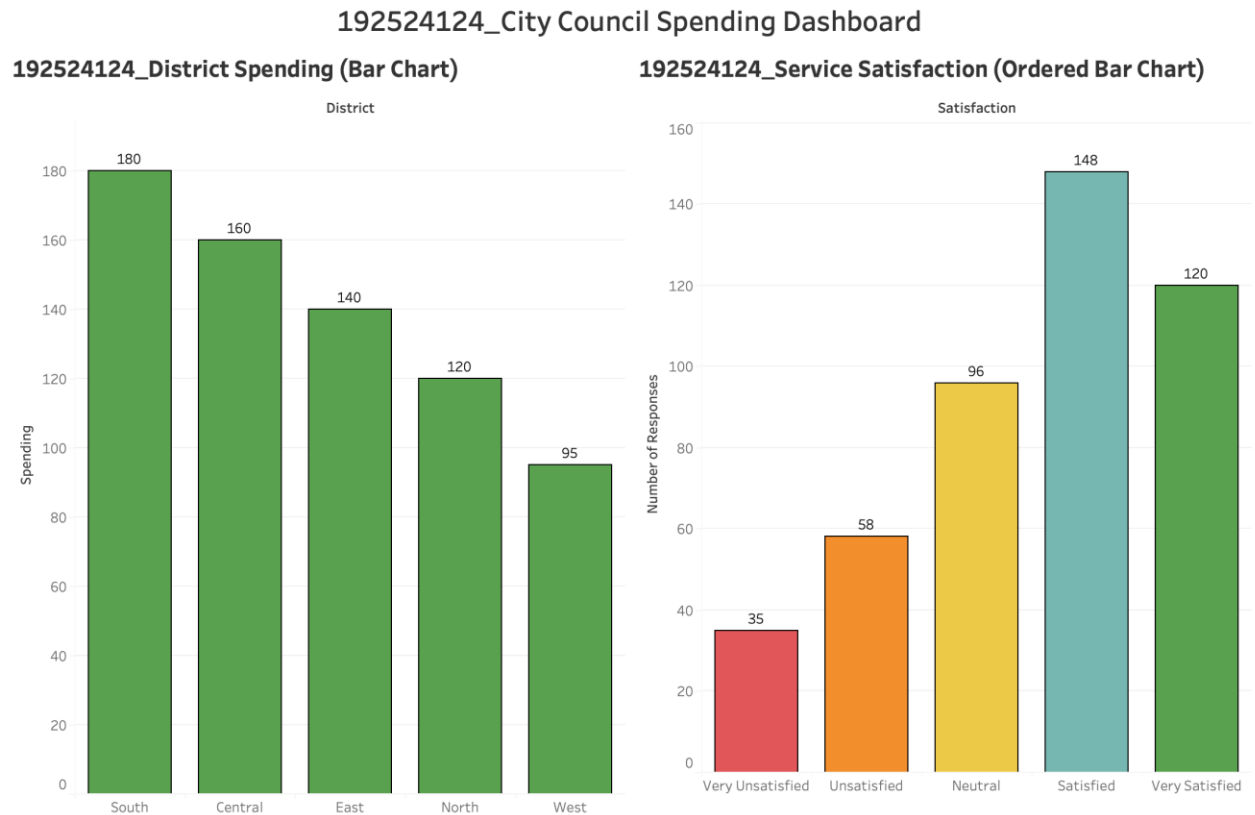
Figure 2c: Year-over-Year Spending Change Diverging Bar Chart

d) Design a combined dashboard using two of the above views, with a narrative takeaway a city council member could act on.

Answer:

The constructed dashboard (Figure 2d) combines the District Spending bar chart from part (a), and the Service Satisfaction ordered bar chart from part (b), placed side by side under a shared 'City Council Spending Dashboard' title. The left panel shows exactly which district received the most budget (South, at 180, the tallest bar), while the right panel shows how residents rate municipal services overall, in the correct ordinal sequence from Very Unsatisfied through to Very Satisfied.

The narrative takeaway written beneath the dashboard notes that the South district received the highest public spending allocation, and that citywide feedback leans more positive than negative overall, while flagging that the sizeable neutral and dissatisfied response counts mean investment effectiveness should continue to be monitored across all districts a direct, actionable prompt for the council to keep reviewing outcomes rather than assuming higher spending alone guarantees satisfaction.



Narrative
The South district received the highest public spending allocation, while overall citizen feedback indicates higher levels of satisfaction than dissatisfaction. However, a noticeable number of neutral and dissatisfied responses suggests that investment effectiveness should be regularly monitored to improve service quality across all districts.

Figure 2d: City Council Spending Dashboard District Spending (Bar) and Service Satisfaction (Ordered Bar)